Mahesh Chitakoti

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Summary

Backend & AI Engineer with 3+ years of software engineering experience building production RAG systems, LLM workflows, and backend services with **Python (FastAPI), TypeScript (Node.js), PostgreSQL, and Redis**. Built LangGraph workflows and Qdrant hybrid search for language learning and reduced recurring LLM inference costs by **35%**.

Technical Skills

Languages & Backend: Python, TypeScript, SQL, FastAPI, Node.js, Express.js, AsyncIO, REST APIs, WebSockets

AI & Retrieval Systems: RAG Pipelines, Agentic Workflows (LangGraph), Qdrant, pgvector, Hybrid Search (Dense + BM25), Semantic Caching, Pydantic Structured Outputs, Gemini APIs, OpenAI APIs

Databases & Cloud Infrastructure: PostgreSQL, Redis, Docker, AWS (EC2, S3), GitHub Actions CI/CD

Experience

Independent Software Developer – Contract & Client Projects
Apr 2026 – Present

Remote

- Engineered an automated document ingestion and data-extraction backend using **FastAPI and Redis Queues**, processing multi-page PDF invoices and client webhooks with structured schema validation.
- Built a domain-specific RAG service using **PostgreSQL (pgvector HNSW indexing)** and semantic caching, cutting average vector retrieval latency from **280 ms to under 185 ms**.
- Containerized and deployed microservice workloads using **Docker on AWS EC2**, provisioning object storage in **S3** and establishing automated linting and test runs via GitHub Actions.

AI & Full Stack Developer – SahioNext Technology Pvt. Ltd.
Feb 2025 – Apr 2026

Remote

- Architected the core automated evaluation engine using **FastAPI and Gemini APIs**, orchestrating multi-step **LangGraph agentic workflows with Pydantic validation** and fallback schemas to enforce JSON output adherence.
- Reduced recurring LLM inference costs by **35%** with exact-match caching and **Redis-backed semantic caching**. Mitigated upstream HTTP 429 errors with token-bucket rate limiting.
- Engineered a hybrid retrieval system (**dense embeddings + BM25**) indexing 10,000+ lesson transcript chunks in **Qdrant**, maintaining under **300ms p95 latency** and improving domain vocabulary recall.
- Developed full-duplex **WebSocket endpoints** for real-time speech evaluation streams, incorporating client state buffering and reconnection handshakes to preserve streaming session persistence.
- Built internal dataset curation and review dashboards using **TypeScript and Node.js**, bridging human-in-the-loop rubric scoring with automated model feedback pipelines.

Full Stack Developer – The DevOps Team
Oct 2023 – Jan 2025

Bangalore, India

- Profiled bottleneck queries and optimized **PostgreSQL indexing** alongside **Redis caching**, reducing median API response times from **800ms to 320ms (60% decrease)**.
- Redesigned candidate search queries using multi-column compound indexes, cutting database query latency and improving recruiter role-matching discovery rates by **22%**.
- Built responsive onboarding and profile workflows in **Next.js**, running A/B experiments on multi-step forms to lift user registration conversion by **18%**.

Projects

WorkBridge – AI Collaboration Platform
Sep 2026 – Present

Python, FastAPI, Node.js, PostgreSQL, pgvector, Redis, Next.js | Live Project

- Architected a decoupled microservices platform pairing a **Node.js API gateway** for session auth and RBAC with an asynchronous **FastAPI service** handling vector calculations and similarity search.
- Built candidate-to-project recommendation matching with **pgvector HNSW indexing**, executing sub-100ms cosine similarity retrieval against structured skill embeddings.
- Benchmarked search endpoints using **k6** under simulated 50-VU load, tuning PostgreSQL connection pooling and Redis session caching to sustain under **150ms p95 latency**.

Certifications

Oracle Agentic AI Certified Foundations Associate – Oracle — Aug 2026
Oracle AI Vector Search Certified Professional – Oracle — Sep 2025
Machine Learning I – Columbia University — Aug 2025
Oracle Cloud Infrastructure Generative AI Professional – Oracle — Aug 2024

Education

Adichunchanagiri Institute of Technology (VTU) — Chikkamagaluru, KA, India
Bachelor of Engineering — *Aug 2023*